

NPLify · P0 Technical Baseline

Deliverable 2 — Ledger Posting Design & Chart of Accounts · Draft v1.0

New XP Technologies Limited

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Status: Built against ERD & Data Model Draft v4.0 and the Project Understanding v1.0 with NPL's review comments. Every worked example below was produced by a small posting engine that refuses any transaction whose debits and credits do not match per currency, so the figures add up by construction.

Audience: NPL Finance and Management, and the New XP engineering team. Written in plain language; the account codes and field names are the ones used in the ERD.

1 · What the ledger is for, in one paragraph

NPL coordinates other people's money. The ledger's job is to answer, at any moment and for any past moment, three questions: **where is the money** (at which partner, in transit, in NPL's own wallet), **whose money is it** (the client's, or NPL's earnings), and **what did each movement cost** (NPL's margin, the partner's cut, bank and network fees, variances). It does this with ordinary double-entry bookkeeping: every movement is a set of debits and credits that balance, nothing is ever edited or deleted, and every balance anyone sees is a sum over those entries.

2 · Rules that every posting follows

1. **One transaction per event.** A collection, a conversion, a disbursement, a return, a confirmation, a bank-fee event, a reroute, a referral or rebate accrual, or a manual adjustment each produce exactly one LEDGER_TRANSACTION that points back at its source record.
2. **Balanced per currency.** Within one transaction the debits equal the credits in each currency separately. A conversion therefore has a USDT part and a EUR part that each balance on their own; no exchange-rate "plug" is ever needed.
3. **Append-only.** A mistake is corrected by a reversing transaction that names the one it reverses, followed by the correct transaction. Reports for any past date still add up.
4. **Every posting carries an owner tag:** *client* (the client's funds) or *company* (NPL's earnings and costs). Client money and company money never sit in the same account.
5. **Every posting carries a cost component:** principal, earnings, partner_cost, bank_fee, network_fee, variance, referral or rounding. Summing a component across postings gives the cost breakdown of any deal, day, partner or client.
6. **Ownership flips from client to company at one event only:** margin recognition at conversion. Money flows the other way (company to client) only when NPL absorbs a fee or tops up a shortfall.

There is no withholding of earnings at source and NPL never advances money to a receiver, so neither has a posting pattern.

7. **Nothing is stored as a balance.** Client balance per partner, group entitlement, offsets, custody and exposure are all sums over postings or deals.

3 · Chart of accounts

Account codes read OWNER.HOLDER.CURRENCY.PURPOSE. The owner is CL for client funds or CO for company. The holder says where the money is or whom it concerns: a partner code such as PT or JETON (a PARTNER_CONFIG, so always within one project), OWN for NPL's own wallet, a receiver, a sender, an introducer, PROJECT for the client as a whole, or NONE for income and expense accounts. One account exists per combination that is actually used; accounts are created on first posting.

Code	Kind	Holder	What the balance means
CL.<PARTNER>.<CCY>.COLLECTED	asset	<PARTNER>	Client money received at this partner, before conversion (in the collection currency).
CL.OWN.<CCY>.HELD	asset	OWN	Client money in NPL's own wallet, only under an approved reroute. Must be zero otherwise.
CL.<PARTNER>.<CCY>.DUE	asset	<PARTNER>	Converted client money held at this partner, waiting to be paid out. One account per project × partner × currency: the client balance per partner.
CL.<PARTNER>.<CCY>.INTRANSIT	asset	<PARTNER>	Client money released by this partner and not yet confirmed by the receiver.
CL.<RECEIVER>.<CCY>.SHORTFALL	asset	<RECEIVER>	Client money delivered short to a receiver and still to be made good on the next settlement.
CL.PROJECT.<CCY>.PAYABLE	liability	PROJECT	What is owed to the client's receivers, in total per currency. The split by receiver group is the GROUP_ENTITLEMENT view, not an account.
CL.<SENDER>.<CCY>.CREDIT	liability	<SENDER>	Owed back to a sender: an overpayment, or a refund waiting to be paid.
CO.<PARTNER>.<CCY>.POOL	asset	<PARTNER>	NPL's earnings resting at this partner (the FRS "PT Comms"). Settled on the partner's fee cycle or netted on a reroute.
CO.OWN.<CCY>.WALLET	asset	OWN	NPL's own money in its wallet, for example dues recovered on a reroute.
CO.<PARTNER>.<CCY>.REBATE_RECEIVABLE	asset	<PARTNER>	Rebate a partner owes NPL on its fees, reconciled monthly.
CO.<INTRODUCER>.<CCY>.REFERRAL_PAYABLE	liability	<INTRODUCER>	Referral commission owed to an introducer.
CO.NONE.<CCY>.EARN_GROSS	income	NONE	Gross margin recognised at conversion. Referral commissions are booked against it, so the balance is NPL's margin after referrals.
CO.NONE.<CCY>.EXP_PARTNER	expense	NONE	Partner cost where the partner states it separately (market + x %). Rebates received are credited here.
CO.NONE.<CCY>.EXP_BANKFEE	expense	NONE	Bank or payout fees NPL absorbs, including gross-ups and shortfall top-ups.

Code	Kind	Holder	What the balance means
C0.NONE.<CCY>.EXP_NETWORK	expense	NONE	Crypto network fees NPL absorbs, for example on a refund.
C0.NONE.<CCY>.VARIANCE	expense	NONE	Difference between the engine's expected conversion and what the partner actually did, and basis variance on a reroute. Can be a gain or a loss.

Debit-normal accounts (assets, expenses) grow with debits; credit-normal accounts (liabilities, income) grow with credits. The ERD's purpose list gains PAYABLE and WALLET as used here and the holder type Project; ADVANCE is not used because NPL never advances money. Rebates are credited to EXP_PARTNER and referral commissions debited to EARN_GROSS, so no extra income or expense accounts are needed.

4 · Reading a posting table

Each worked example shows one transaction as a table. *Debit* and *Credit* are the amounts; *Owner* is the ownership tag; *Cost component* is the breakdown tag; *Why* says in plain words what that line records. The running example uses project Evo, sender Ayush, receiver Sud (receiver group A, the default group) with entities NewXP Entity and ReferScout Entity, partner PT (Ali, a disclosed-rate partner whose margin is inside its rate) and partner Jeton (a market + 0.50 % partner that states its fee separately). Rates: market 0.9000 EUR per USDT, PT 0.8950 EUR per USDT. Fee structure: basis partner, fixed, 1.00 % in total, 0.40 % sender share and 0.60 % receiver share. The arithmetic behind the amounts is in the Calculation Specification; the headline figures for deal 1 are:

Figure	Deal 1 (10,000 USDT at PT)	How
Sender rate	0.89142	$0.8950 \times (1 - 0.40 \%)$
Gross out	8,914.20 EUR	$10,000 \times \text{sender rate}$
Receiver share	53.49 EUR	$\text{gross out} \times 0.60 \%$
Client net (entitlement)	8,860.71 EUR	$\text{gross out} - \text{receiver share}$
Converted at PT	8,950.00 EUR	$10,000 \times 0.8950$
NPL earnings	89.29 EUR	$\text{converted} - \text{client net} (= \text{sender share } 35.80 + \text{receiver share } 53.49)$
Paid out in whole euros / dust	8,860.00 / 0.71 EUR	whole-unit rule; the dust stays client money

5 · Pattern A — direct collection, conversion and accumulation

The everyday case: the sender pays, the partner converts, the client balance at that partner grows until it is disbursed. Collect-first and quote-first deals post identically; only the moment pricing is fixed differs.

T1 · Collection of deal 1: 10,000 USDT arrive at PT's wallet source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.USDT.COLLECTED	10,000.00		USDT	client	principal	client money now sits at PT, unconverted
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	NPL owes this onward on the client's behalf

T2 · Conversion of deal 1 at PT: 10,000 USDT → 8,950.00 EUR at 0.8950 source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		USDT	client	principal	the USDT obligation is settled by the conversion
CL.PT.USDT.COLLECTED		10,000.00	USDT	client	principal	the USDT left the collected position
CL.PT.EUR.DUE	8,860.00		EUR	client	principal	client net in whole euros, waiting to be paid
CL.PT.EUR.DUE	0.71		EUR	client	rounding	the cents below one euro stay client money (Q8)
CO.PT.EUR.POOL	89.29		EUR	company	earnings	NPL's margin, physically still at PT
CL.PROJECT.EUR.PAYABLE		8,860.71	EUR	client	principal	owed to the receiver group, now in EUR
CO.NONE.EUR.EARN_GROSS		89.29	EUR	company	earnings	margin recognised: the only client → company event

Ownership changes here and only here: 89.29 EUR becomes company money. The partner's own cost is inside its rate (disclosed rate), so no partner-cost posting is needed; the inferred cost against market is 50.00 EUR and is kept as information only.

Balances after deal 1

Account	Ccy	Balance	Meaning
CL.PT.USDT.COLLECTED	USDT	0.00	nothing left unconverted
CL.PT.EUR.DUE	EUR	8,860.71	client balance at PT (the per-partner balance)
CL.PROJECT.EUR.PAYABLE	EUR	8,860.71	owed to receivers; equals group entitlement
CO.PT.EUR.POOL	EUR	89.29	NPL's margin resting at PT
CO.NONE.EUR.EARN_GROSS	EUR	89.29	margin recognised to date

6 · Pattern B — part collection

A sender may pay in parts (80K today, 20K tomorrow). Each part is posted when it is verified. The conversion is one transaction once the deal is fully collected, unless NPL instructs the partner to convert a part early, in which case each converted part posts like T5 for its own amount.

T3 • Collection of deal 2, part 1 of 2: 4,000 USDT source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.USDT.COLLECTED	4,000.00		USDT	client	principal	first part
CL.PROJECT.USDT.PAYABLE		4,000.00	USDT	client	principal	

T4 • Collection of deal 2, part 2 of 2: 1,000 USDT (next day) source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.USDT.COLLECTED	1,000.00		USDT	client	principal	second part completes the deal
CL.PROJECT.USDT.PAYABLE		1,000.00	USDT	client	principal	

T5 • Conversion of deal 2 at PT: 5,000 USDT → 4,475.00 EUR source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	5,000.00		USDT	client	principal	
CL.PT.USDT.COLLECTED		5,000.00	USDT	client	principal	
CL.PT.EUR.DUE	4,430.00		EUR	client	principal	
CL.PT.EUR.DUE	0.36		EUR	client	rounding	
CO.PT.EUR.POOL	44.64		EUR	company	earnings	
CL.PROJECT.EUR.PAYABLE		4,430.36	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		44.64	EUR	company	earnings	

Parts are posted as they arrive; the conversion is one transaction once the deal is fully collected.

7 • Pattern C — accumulated and instant settlement, bank fees, shortfalls

There is one settlement model. "Instant" simply means the balance is disbursed as soon as one deal has accumulated; "accumulated" means several deals first. The postings are the same. Disbursements pay whole units; the cents below one unit stay in the client balance and go out when they add up to a unit.

T6 • Disbursement from PT: 13,291.00 EUR in two lines (NewXP Entity 9,000.00, ReferScout Entity 4,291.00) source: Disbursement

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.EUR.INTRANSIT	9,000.00		EUR	client	principal	line 1 released by PT
CL.PT.EUR.INTRANSIT	4,291.00		EUR	client	principal	line 2 released by PT
CL.PT.EUR.DUE		13,291.00	EUR	client	principal	whole euros leave DUE; 0.07 EUR of rounding dust stays

Disburse to less than one unit: the client balance at PT drops from 13,291.07 to 0.07 EUR. Both lines pay entities of the same receiver, so no offset arises.

T7 • Confirmation, line 1 full: NewXP Entity received 9,000.00 EUR source: Confirmation

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.EUR.PAYABLE	9,000.00		EUR	client	principal	the obligation is discharged
CL.PT.EUR.INTRANSIT		9,000.00	EUR	client	principal	the money reached the receiver

T8 • Confirmation, line 2 short: ReferScout received 4,241.00 EUR, bank took 50.00 (treatment carry_forward) source: Confirmation

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.EUR.PAYABLE	4,241.00		EUR	client	principal	delivered part
CL.SUD.EUR.SHORTFALL	50.00		EUR	client	bank_fee	still owed to the receiver, topped up next settlement
CL.PT.EUR.INTRANSIT		4,291.00	EUR	client	principal	nothing is left in transit

With treatment *waived* the second line would instead be Dr CL.PROJECT.EUR.PAYABLE 50 (the client bears the fee and its claim falls by 50). With *absorbed* the fee is expected up front: see T9.

T9 • Shortfall top-up funded from NPL's pool at PT (BANK_FEE_EVENT applied to the next disbursement) source: BankFeeEvent

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.NONE.EUR.EXP_BANKFEE	50.00		EUR	company	bank_fee	NPL bears the cost
CO.PT.EUR.POOL		50.00	EUR	company	bank_fee	paid out of NPL's earnings resting at PT
CL.PT.EUR.DUE	50.00		EUR	client	bank_fee	the 50 EUR is client money again, in balance at PT
CL.SUD.EUR.SHORTFALL		50.00	EUR	client	bank_fee	the shortfall is cleared

The 50 EUR now sits in DUE and goes out with the next disbursement like any other balance. Company → client transfers like this one are the only way company money becomes client money.

Balances after settlement of deals 1 and 2

Account	Ccy	Balance	Meaning
CL.PT.EUR.DUE	EUR	50.07	rounding dust, plus the 50 EUR top-up waiting for the next disbursement
CL.PT.EUR.INTRANSIT	EUR	0.00	nothing in transit
CL.PROJECT.EUR.PAYABLE	EUR	50.07	still owed to receivers: the dust and the top-up
CL.SUD.EUR.SHORTFALL	EUR	0.00	cleared
CO.PT.EUR.POOL	EUR	83.93	NPL's earnings at PT after funding the top-up
CO.NONE.EUR.EXP_BANKFEE	EUR	50.00	bank fees NPL has borne

The three bank-fee treatments on a 50 EUR fee. *Waived*: the client bears it; the confirmation debits CL.PROJECT.EUR.PAYABLE for the full released amount and the client's claim simply ends 50 lower. *Carry-forward*: T8 and T9, the fee becomes a shortfall that NPL funds from its pool and adds to the next disbursement. *Absorbed* (gross-up): NPL releases 50 more than the client net so the receiver nets the right amount; at release Dr CO.NONE.EUR.EXP_BANKFEE 50 / Cr CO.PT.EUR.POOL 50 is posted alongside the client's in-transit line, and a BANK_FEE_EVENT later trues up expected against actual. The project sets the default; one settlement can override it with an approved FEE_DECISION.

8 · Pattern D — reroute through NPL's own wallet, with dues recovered

This is the only time NPL holds client money. When the intended partner cannot take the volume, an approved REROUTE collects into NPL's own wallet and forwards to an alternate partner. In v4.0 there is **no withholding of earnings at source**: NPL's margin is captured when the alternate partner converts, exactly as in Pattern A. What a reroute can do is **recover dues**: net what the alternate partner already owes NPL against the money being forwarded.

T10 · Deal 3, PT unavailable: 10,000 USDT collected into NPL's own wallet under an approved reroute source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.OWN.USDT.HELD	10,000.00		USDT	client	principal	client money in custody, with deal, reason and age
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	

Only an approved REROUTE may debit CL.OWN.*.HELD (invariant 3). The custody aging view shows this balance until it is forwarded.

T11 · Forward to Jeton with recover dues: 10,000 USDT of client money credited at Jeton, 30.00 USDT of NPL's dues netted source: Reroute

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.JETON.USDT.COLLECTED	10,000.00		USDT	client	principal	Jeton credits the full client amount
CL.OWN.USDT.HELD		10,000.00	USDT	client	principal	custody ends
CO.OWN.USDT.WALLET	30.00		USDT	company	earnings	NPL keeps 30 USDT in its wallet instead of sending it
CO.JETON.USDT.POOL		30.00	USDT	company	earnings	Jeton's debt to NPL falls by the same 30 USDT

NPL physically sends 9,970.00 USDT; Jeton agrees to treat it as 10,000 of client money because it owed NPL 30.00 USDT of earnings from earlier deals. The recovery is an asset swap (receivable → cash), never income, and every deduction names the deal it recovers. **No earnings are withheld from the client amount**: NPL's margin is captured when Jeton converts (T12).

T12 • Conversion of deal 3 at Jeton (market + 0.50 %): 10,000 USDT → 9,000.00 EUR at market, Jeton keeps 45.00 source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		USDT	client	principal	
CL.JETON.USDT.COLLECTED		10,000.00	USDT	client	principal	
CL.JETON.EUR.DUE	8,910.00		EUR	client	principal	client net, whole euros
CL.JETON.EUR.DUE	0.22		EUR	client	rounding	
CO.JETON.EUR.POOL	44.78		EUR	company	earnings	NPL's net margin resting at Jeton
CO.NONE.EUR.EXP_PARTNER	45.00		EUR	company	partner_cost	Jeton's stated fee, deducted at transaction level
CL.PROJECT.EUR.PAYABLE		8,910.22	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		89.78	EUR	company	earnings	gross margin before partner cost

Total cost to the client = partner cost 45.00 + NPL earnings 44.78 = 89.78 EUR, the gross margin. For a disclosed-rate partner (T2) the partner cost is inside the rate and no EXP_PARTNER line exists.

Balances after deal 3

Account	Ccy	Balance	Meaning
CL.OWN.USDT.HELD	USDT	0.00	custody is empty again
CO.OWN.USDT.WALLET	USDT	30.00	dues recovered into NPL's wallet
CL.JETON.EUR.DUE	EUR	8,910.22	client balance at Jeton
CO.JETON.EUR.POOL	EUR	44.78	NPL's margin resting at Jeton
CO.NONE.EUR.EXP_PARTNER	EUR	45.00	Jeton's stated fee, before any rebate

A recovery is only allowed up to the outstanding receivable at that partner (ERD invariant 15): in real operation CO.JETON.USDT.POOL is positive from earlier deals before T11 and 30 USDT smaller after it. This document's examples start from an empty ledger, so that account is not shown above.

9 • Pattern E — cancellations and reversals

- **Void before any money moved:** a state change with a reason; no postings.
- **Erroneous posting:** never edited. A reversing transaction mirrors it and names it; the correct transaction follows (T13, T14).
- **Void after collection:** the money goes back to the sender (network-fee policy decides who bears the fee), is kept as sender credit, or is held; Management approves the disposition (T15).
- **Rejected payout:** the released line comes back to the partner minus the bank's fee; the line is reversed into the client balance and the fee is booked (T18). The money then waits for a later payout or funds a return leg (T19).

T13 • Erroneous posting: a collection of 2,000 USDT was booked against the wrong deal (as posted) source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.USDT.COLLECTED	2,000.00		USDT	client	principal	booked in error
CL.PROJECT.USDT.PAYABLE		2,000.00	USDT	client	principal	

T14 • Reversal of T13 (LEDGER_TRANSACTION.reversal_of_id = T13), then T13 is re-posted correctly source: Adjustment

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	2,000.00		USDT	client	principal	mirror image of T13
CL.PT.USDT.COLLECTED		2,000.00	USDT	client	principal	

Nothing is edited or deleted: the wrong transaction stays, the reversal points at it, and the correct transaction follows. Any report at any past moment still adds up.

T15a • Collection of deal 5: 3,000 USDT arrive at PT (later voided) source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.USDT.COLLECTED	3,000.00		USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		3,000.00	USDT	client	principal	

T15 • Deal voided after collection, refund to the sender in USDT; network fee 2.00 USDT charged back to the sender (project policy) source: Adjustment

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		USDT	client	principal	the obligation ends
CL.PT.USDT.COLLECTED		2,998.00	USDT	client	principal	returned to the sender's wallet
CL.PT.USDT.COLLECTED		2.00	USDT	client	network_fee	paid to the network; the sender bears it

If the project policy (or a FEE_DECISION for this transaction) says NPL absorbs the fee, the last line becomes Dr CO.NONE.USDT.EXP_NETWORK 2.00 / Cr CO.PT.USDT.P00L 2.00 and the sender receives the full 3,000. Management approves the disposition of any post-money void.

10 · Pattern F — rejected payout and return leg

Deal 4 converts 2,000 USDT to GBP at PT, is paid out, and bounces at the receiving bank. The failed line is reversed first, which restores the receiver group's entitlement. NPL can then leave the GBP in balance for a later payout to any receiver in the same group (another group would be an approved offset), or open a **return leg** on the same deal group that converts back and pays the sender's return receiver.

T16 · Deal 4 collected and converted at PT: 2,000 USDT → 1,550.00 GBP (collection and conversion shown together) source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.USDT.COLLECTED	2,000.00		USDT	client	principal	collection
CL.PROJECT.USDT.PAYABLE		2,000.00	USDT	client	principal	collection
CL.PROJECT.USDT.PAYABLE	2,000.00		USDT	client	principal	conversion, USDT leg
CL.PT.USDT.COLLECTED		2,000.00	USDT	client	principal	conversion, USDT leg
CL.PT.GBP.DUE	1,534.00		GBP	client	principal	
CL.PT.GBP.DUE	0.54		GBP	client	rounding	
CO.PT.GBP.POOL	15.46		GBP	company	earnings	
CL.PROJECT.GBP.PAYABLE		1,534.54	GBP	client	principal	
CO.NONE.GBP.EARN_GROSS		15.46	GBP	company	earnings	

T17 · Disbursement of deal 4: 1,534.00 GBP released to the receiver's bank source: Disbursement

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.GBP.INTRANSIT	1,534.00		GBP	client	principal	
CL.PT.GBP.DUE		1,534.00	GBP	client	principal	

T18 · The receiving bank rejects the transfer: 1,519.00 GBP come back to PT, the bank kept 15.00 (DISBURSEMENT_RETURN; line outcome = returned) source: DisbursementReturn

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.GBP.DUE	1,519.00		GBP	client	principal	back in the client balance at PT
CO.NONE.GBP.EXP_BANKFEE	15.00		GBP	company	bank_fee	bounce fee absorbed by NPL (treatment absorbed)
CL.PT.GBP.INTRANSIT		1,534.00	GBP	client	principal	nothing is in transit any more
CL.PT.GBP.DUE	15.00		GBP	client	bank_fee	NPL makes the client whole
CO.PT.GBP.POOL		15.00	GBP	company	bank_fee	funded from NPL's pool at PT

The failed line is reversed before anything else happens, so the receiver group's entitlement is restored. The money can now stay in balance for a later payout (option B) or fund a return leg (option A, T19).

T19 · Return leg: 1,534.00 GBP converted back to 1,971.19 USDT at PT (rate 1.285); markup waived by FEE_DECISION source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.GBP.PAYABLE	1,534.00		GBP	client	principal	GBP obligation ends
CL.PT.GBP.DUE		1,534.00	GBP	client	principal	
CL.PT.USDT.DUE	1,971.19		USDT	client	principal	USDT owed to the sender's return receiver
CL.PROJECT.USDT.PAYABLE		1,971.19	USDT	client	principal	

NPL's margin from T16 (15.46 GBP) stays in the pool: it was captured at the first conversion and is never at risk from a return. The round trip's rate loss falls on the client's funds. Had the FEE_DECISION applied a markup, part of the 1,971.19 USDT would have been posted to CO.PT.USDT.POOL and EARN_GROSS exactly as in T2. The payout of 1,971.19 USDT to the sender's wallet then follows T17 and T7.

11 · Pattern G — inter-recipient offsets

An offset is **not** a ledger event. When a disbursement line pays a receiver in another receiver group than the one entitled to the money, the postings are exactly T6 and T7; the ledger sees client money leaving the client balance and reaching a receiver. What changes is the GROUP_ENTITLEMENT view: the paid group now shows less entitlement than it should (or a negative figure), the entitled group shows more, and the two differences are equal. The offset clears as later collections for the entitled group are converted and later payouts favour it. Within one group, paying any receiver or entity is ordinary settlement and nothing is tracked beyond an informational tally. Offsets across groups carry Management approval and tighter aging.

	Group A entitled	Group B entitled	Client balance at PT	Comment
After converting 9,000 EUR for A and 6,000 EUR for B	9,000.00	6,000.00	15,000.00	entitlements sum to the balance (invariant 7)
Pay A 4,000	5,000.00	6,000.00	11,000.00	ordinary settlement
Pay B 8,000 (B's accounts needed it while A's were blocked)	5,000.00	-2,000.00	3,000.00	B holds 2,000 of A's flow: the offset, approved by Management
Convert 2,000 more for A, then pay A 5,000	2,000.00	-2,000.00	0.00	balance empty; A is still owed 2,000 by B until B's next collections clear it

Advances. NPL never pays a receiver before the client's money has arrived, so there is no advance account and no advance posting pattern in this design.

12 · Pattern H — balance conversion

NPL may convert a client's held balance from one currency to another to fund a payout, for example 5,000 SGD to USDT at PT. It is not a deal: there is no sender, receiver or collection, and Finance requests and approves it. The posting moves the client's asset and obligation from one currency to the other; a markup, if a FEE_DECISION applies one, posts to the pool and to gross earnings as in T2.

T20a • Opening position: 5,000 SGD of converted client balance at PT from earlier deals (summarised) source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.SGD.DUE	5,000.00		SGD	client	principal	balance built up by earlier SGD deals
CL.PROJECT.SGD.PAYABLE		5,000.00	SGD	client	principal	

T20 • Balance conversion at PT: 5,000 SGD of client balance → USDT at 0.7400, no markup (serves Group B) source: BalanceConversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.SGD.PAYABLE	5,000.00		SGD	client	principal	SGD obligation ends
CL.PT.SGD.DUE		5,000.00	SGD	client	principal	SGD balance leaves
CL.PT.USDT.DUE	3,700.00		USDT	client	principal	USDT balance arrives, whole units
CL.PROJECT.USDT.PAYABLE		3,700.00	USDT	client	principal	same obligation, new currency

No sender, receiver or collection: Finance requests and approves. ENTITLEMENT_REATTRIBUTION moves Group B's SGD entitlement into USDT; if Group B held less than 5,000 SGD of entitlement, the excess is another group's flow and is booked as an approved offset.

13 • Pattern I — referral commissions and partner rebates

T21 • Referral commission on deal 3: introducer John is owed 25 % of NPL's earnings = 11.20 EUR source: Referral

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.NONE.EUR.EARN_GROSS	11.20		EUR	company	referral	reduces NPL's margin
CO.JOHN.EUR.REFERRAL_PAYABLE		11.20	EUR	company	referral	owed to the introducer

A fixed-percentage introducer (1 % of the transaction inside the sender-facing markup) is priced into the sender rate instead; the posting is the same shape with the amount from the calculation specification.

T22 • Partner rebate on deal 3: Jeton returns 32 % of its fee = 14.40 EUR (reconciled monthly) source: Rebate

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CO.JETON.EUR.REBATE_RECEIVABLE	14.40		EUR	company	partner_cost	expected from Jeton
CO.NONE.EUR.EXP_PARTNER		14.40	EUR	company	partner_cost	net partner cost falls

When Jeton pays (usually in crypto) the receivable is cleared against NPL's wallet or pool.

14 · Pattern J — overpayment and partner variance

T23 · Overpayment: a sender sends 10,050 USDT against a 10,000 USDT deal; 50 USDT becomes sender credit source: Collection

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PT.USDT.COLLECTED	10,050.00		USDT	client	principal	all of it is client money at PT
CL.PROJECT.USDT.PAYABLE		10,000.00	USDT	client	principal	the deal
CL.AYUSH.USDT.CREDIT		50.00	USDT	client	principal	owed back to the sender, or applied to the next deal

T24 · Conversion where the partner calculates: expected 2,685.00 EUR, partner booked 2,683.50 EUR; variance 1.50 EUR source: Conversion

Account	Debit	Credit	Ccy	Owner	Cost component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		USDT	client	principal	
CL.PT.USDT.COLLECTED		3,000.00	USDT	client	principal	
CL.PT.EUR.DUE	2,658.00		EUR	client	principal	client net is unchanged: the sender's price is fixed
CL.PT.EUR.DUE	0.21		EUR	client	rounding	
CO.PT.EUR.POOL	25.29		EUR	company	earnings	margin actually left at PT
CO.NONE.EUR.VARIANCE	1.50		EUR	company	variance	what the partner did worse than expected
CL.PROJECT.EUR.PAYABLE		2,658.21	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		26.79	EUR	company	earnings	margin as priced

The sender's stamped price fixes the client net, so a partner that converts worse than expected costs NPL, not the client. An exception opens when the variance is beyond the project's tolerance; the deal does not wait.

15 · Daily identities Finance checks

Identity	Check
Client due per partner = Σ group entitlements	For each currency: client assets equal client liabilities, Σ COLLECTED + Σ HELD + Σ DUE + Σ INTRANSIT + Σ SHORTFALL = CL.PROJECT. <CCY>.PAYABLE + Σ CREDIT; and PAYABLE = Σ GROUP_ENTITLEMENT (ERD invariant 7).
Pool per partner = partner-stated comms balance $\pm$ logged difference	CO.<PARTNER>.<CCY>.POOL agrees to the partner's statement of what it holds for NPL.
Exposure $\leq$ policy	Σ DUE + Σ INTRANSIT at a partner is within the project's exposure ceiling; under disburse_policy to_zero, DUE after each release is below one unit.
Own-wallet holdings = 0 unless an open reroute	CL.OWN.<CCY>.HELD is zero except for rows the custody aging view can name (deal, reason, age).
Each confirmation = its in-transit amount exactly	Confirmed + shortfall + fees booked = the amount released.
Ownership flips once	The only client $\rightarrow$ company postings are EARN_GROSS recognitions at conversion. Company $\rightarrow$ client postings exist only for fees NPL absorbs.
Every balance shown is a sum	No screen or report shows a stored figure; all are sums over postings or deals (ERD invariant 10).

16 · Transaction sources and what they post

Source event	Posts to	Owner tags	Cost components
Collection	COLLECTED or HELD, PAYABLE, CREDIT	client	principal
Conversion (deal leg)	COLLECTED/PAYABLE in the in-currency; DUE, PAYABLE, POOL, EARN_GROSS, EXP_PARTNER, VARIANCE in the out-currency	client and company	principal, rounding, earnings, partner_cost, variance
BalanceConversion	DUE and PAYABLE in both currencies; POOL and EARN_GROSS if markup applied	client (and company)	principal, rounding, earnings
Disbursement	INTRANSIT, DUE; EXP_BANKFEE and POOL when a fee is absorbed up front	client (and company)	principal, bank_fee
DisbursementReturn	DUE, INTRANSIT; EXP_BANKFEE and POOL for the bounce fee	client and company	principal, bank_fee
Confirmation	PAYABLE, INTRANSIT, SHORTFALL	client	principal, bank_fee
BankFeeEvent	EXP_BANKFEE, POOL, DUE, SHORTFALL	company and client	bank_fee
Reroute	HELD, COLLECTED at the alternate partner; WALLET and POOL for recovered dues	client and company	principal, earnings
Referral	EARN_GROSS, REFERRAL_PAYABLE	company	referral
Rebate	REBATE_RECEIVABLE, EXP_PARTNER	company	partner_cost
Adjustment	any, always as a reversal plus a correct re-posting, or a Management-approved disposition	as reversed	as reversed

17 · Who sees what

Operations sees operational fields, sender-facing prices and partner rates as entered, but never a ledger balance, a margin, a pool, a cost breakdown or a variance, on any screen, export, message or report.

Finance sees all postings and balances and owns reconciliation. Management additionally approves settlements above limit, cross-group payouts, shortfall top-ups, contingency routes, own-wallet outbound transfers, balance conversions, dues recovery and post-money voids. Approver is never the initiator, and every controlled action is audited with actor, time, reason and record version.

Draft v1.0 — for review with NPL. Figures are worked examples, not NPL data. The account purposes PAYABLE and WALLET and the holder type Project are additions to the ERD v4.0 enumerations and will be carried into the next ERD revision.